MAT21-2

Probability & Statistics I

Discrete probability, distributions, estimation, and Bayesian reasoning

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The mathematical language of uncertainty for discrete random phenomena.

Albert School — 34 hours

Contents

1 Counting

Definition 1 A **permutation** of a set of n distinct objects is an ordering of all of them. The number of permutations is

$$n! = n(n-1)(n-2)...2 \cdot 1, \quad 0! = 1.$$

Definition 2 An **arrangement** (or k-permutation) of n distinct objects is an ordered selection of k of them $(0 \le k \le n)$. Their number is

$$A_n^k = \frac{n!}{(n-k)!} = n(n-1)...(n-k+1).$$

Definition 3 A **combination** of n distinct objects taken k at a time is an unordered selection of k of them. Their number is the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Definition 4 For non-negative integers $k_1, ..., k_r$ with $k_1 + ... + k_r = n$, the **multinomial coefficient**

$$\binom{n}{k_1, ..., k_r} = \frac{n!}{k_1!k_2!...k_r!}$$

counts the ways to partition n distinct objects into r labelled groups of sizes $k_1, ..., k_r$.

2 Finite probability spaces

Definition 5 A **finite probability space** is a pair (Ω, P) where:

- the **sample space** Ω is a finite non-empty set, whose elements are the **outcomes**;
- an **event** is a subset $A \subseteq \Omega$;
- the **probability measure** P assigns to each event a number $P(A) \in [0, 1]$ satisfying the **axioms**:
  1. $P(\Omega) = 1$;
  2. for disjoint events A and B (i.e. $A \cap B = \varnothing$), $P(A \cup B) = P(A) + P(B)$.

From the axioms follow $P(\varnothing) = 0$, $P(A^c) = 1 - P(A)$, and more generally $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Definition 6 The space is **equiprobable** (uniform) when every outcome has the same probability. Then for any event A,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}.$$

3 Conditional probability

Definition 7 Let A, B be events with $P(B) > 0$. The **conditional probability** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Proposition 8 (Multiplication rule) For events with positive probability,

$$P(A \cap B) = P(B)P(A \mid B) = P(A)P(B \mid A).$$

More generally,

$$P(A_1 \cap \ldots \cap A_n) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 \cap A_2) \ldots P(A_n \mid A_1 \cap \ldots \cap A_{n-1}).$$

Definition 9 Events $B_1, \ldots, B_n$ form a **partition** of Ω if they are pairwise disjoint and their union is Ω.

Theorem 10 (Total probability formula) If $B_1, \ldots, B_n$ is a partition of Ω with each $P(B_i) > 0$, then for any event A,

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Theorem 11 (Bayes' theorem) If $B_1, \ldots, B_n$ is a partition of Ω with each $P(B_i) > 0$ and $P(A) > 0$, then for each j,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}.$$

4 Independence

Definition 12 Two events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

Equivalently, when $P(B) > 0$, if $P(A \mid B) = P(A)$.

Definition 13 Events $A_1, \ldots, A_n$ are **mutually independent** if for every subset $I \subseteq \{1, \ldots, n\}$,

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i).$$

They are **pairwise independent** if this holds for every pair. Mutual independence implies pairwise independence; the converse is false.

5 Discrete random variables

Definition 14 A **discrete random variable** X on (Ω, P) is a map $X : \Omega \to \mathbb{R}$ taking values in a finite or countable set $X(\Omega)$. Its **probability mass function** (PMF) is

$$p_X(x) = P(X = x), \quad x \in X(\Omega),$$

satisfying $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$.

Definition 15 The **cumulative distribution function** (CDF) of X is

$$F_X(x) = P(X \leq x) = \sum_{t \leq x} p_X(t).$$

It is non-decreasing, right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$.

5.1 Expectation and variance

Definition 16 The **expectation** (mean) of a discrete random variable X is

$$\mathbb{E}[X] = \sum_{x \in X(\Omega)} x \, p_X(x),$$

when this sum converges absolutely. For a function g,

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).$$

Definition 17 The **variance** of X is

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] \geq 0,$$

and its **standard deviation** is $\sqrt{\mathrm{Var}(X)}$.

Proposition 18 (König–Huygens formula)

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proposition 19 (Linearity of expectation) For random variables X, Y and scalars a, b,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

whether or not X and Y are independent.

Proposition 20 For a scalar a, $\mathrm{Var}(aX + b) = a^2 \, \mathrm{Var}(X)$. If $X_1, ..., X_n$ are **independent**, then

$$\mathrm{Var}(X_1 + ... + X_n) = \mathrm{Var}(X_1) + ... + \mathrm{Var}(X_n).$$

Definition 21 Random variables X and Y are **independent** if for all x, y,

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

6 Classic discrete distributions

Each distribution below is determined by a **modelling assumption**; the PMF is derived from that assumption.

6.1 Bernoulli

Definition 22 A **Bernoulli** trial has two outcomes, **success** (value 1) with probability p and **failure** (value 0) with probability $1 - p$. Then $X \sim \mathrm{Ber}(p)$ has

$$P(X = 1) = p, \quad P(X = 0) = 1 - p,$$

with $\mathbb{E}[X] = p$ and $\mathrm{Var}(X) = p(1 - p)$.

6.2 Binomial

Definition 23 Count the number of successes in n independent Bernoulli trials each with success probability p. The result $X \sim \mathrm{Bin}(n, p)$ has

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, ..., n,$$

with $\mathbb{E}[X] = np$ and $\mathrm{Var}(X) = np(1-p)$.

The binomial is the sum of n independent $\mathrm{Ber}(p)$ variables, which gives its mean and variance by linearity and additivity of variance.

6.3 Geometric

Definition 24 Count the number of independent $\mathrm{Ber}(p)$ trials up to and including the first success. Then $X \sim \mathrm{Geom}(p)$ has

$$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, 3, ...,$$

with $\mathbb{E}[X] = \frac{1}{p}$ and $\mathrm{Var}(X) = \frac{1-p}{p^2}$.

6.4 Poisson

Definition 25 The **Poisson** distribution with parameter $\lambda > 0$ models the number of occurrences of a rare event in a fixed interval. Then $X \sim \mathrm{Poi}(\lambda)$ has

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, ...,$$

with $\mathbb{E}[X] = \lambda$ and $\mathrm{Var}(X) = \lambda$.

The Poisson arises as the limit of $\mathrm{Bin}(n, p)$ when $n \to \infty$ and $p \to 0$ with $np \to \lambda$.

6.5 Model construction

Selecting a distribution for a problem means translating the word problem into a sample space, the relevant events, and the random variable, then matching the modelling assumption (number of fixed trials, waiting time to first success, rare count, ...) to the distribution above — reasoning over recognition.

7 Introduction to estimation

Definition 26 Given data $X_1, ..., X_n$ drawn from a distribution depending on an unknown parameter θ, an **estimator** $\hat{\theta} = \hat{\theta}(X_1, ..., X_n)$ is a random variable used to estimate θ.

Definition 27 The **bias** of an estimator is

$$\mathrm{bias}\left(\hat{\theta}\right) = \mathbb{E}\left[\hat{\theta}\right] - \theta.$$

The estimator is **unbiased** if $\mathrm{bias}\left(\hat{\theta}\right) = 0$.

Definition 28 The **mean squared error** of $\hat{\theta}$ is

$$\mathrm{MSE}\left(\hat{\theta}\right) = \mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right] = \mathrm{Var}\left(\hat{\theta}\right) + \mathrm{bias}\left(\hat{\theta}\right)^2.$$

8 Maximum likelihood estimation

Definition 29 Given observed data $x_1, ..., x_n$ from a discrete distribution with PMF $p(x; \theta)$, the **likelihood function** is

$$L(\theta) = \prod_{i=1}^{n} p(x_i; \theta),$$

and the **log-likelihood** is $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln p(x_i; \theta)$. The **maximum likelihood estimator** (MLE) $\hat{\theta}$ maximises $L(\theta)$ (equivalently $\ell(\theta)$).

Worked example — Bernoulli. For $x_1, ..., x_n \in \{0, 1\}$ from $\mathrm{Ber}(p)$, with $s = \sum_i x_i$,

$$\ell(p) = s \ln p + (n - s) \ln(1 - p), \quad \ell'(p) = \frac{s}{p} - \frac{n - s}{1 - p} = 0,$$

giving $\hat{p} = \frac{s}{n} = \overline{x}$. This estimator is unbiased.

Worked example — Binomial. For a single observation $x \sim \mathrm{Bin}(n, p)$ with n known, $\hat{p} = \frac{x}{n}$, unbiased.

Worked example — Poisson. For $x_1, ..., x_n$ from $\mathrm{Poi}(\lambda)$,

$$\ell(\lambda) = -n\lambda + \left(\sum_i x_i\right) \ln \lambda - \sum_i \ln(x_i!), \quad \ell'(\lambda) = -n + \frac{\sum_i x_i}{\lambda} = 0,$$

giving $\hat{\lambda} = \overline{x}$, unbiased.

9 Bayesian updating

Definition 30 In **Bayesian** inference the unknown parameter θ is treated as random with a **prior** distribution $\pi(\theta)$ encoding beliefs before data. Given data D with **likelihood** $P(D \mid \theta)$, the **posterior** distribution is obtained by Bayes' theorem:

$$\pi(\theta \mid D) = \frac{P(D \mid \theta)\, \pi(\theta)}{P(D)}, \quad P(D) = \sum_{\theta} P(D \mid \theta) \pi(\theta).$$

The posterior combines prior and data. As evidence accumulates the likelihood dominates and the posterior concentrates regardless of the prior; with little data the prior matters more.

Worked example — Disease testing. A disease has prevalence $P(D) = 0.01$. A test has sensitivity $P(+ \mid D) = 0.99$ and false-positive rate $P(+ \mid D^c) = 0.05$. By Bayes' theorem,

$$P(D \mid +) = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} \approx 0.167.$$

> Despite the positive test, the posterior probability of disease stays low because the prior prevalence is small.

> **Worked example — Quality control.** A batch is **good** (defect rate 0.01) with prior 0.9 or **bad** (defect rate 0.1) with prior 0.1. Observing a defective item updates the posterior toward **bad**; each further defective item shifts it further, illustrating how repeated evidence overwhelms the prior.